Muhammad Kashif

PhD Researcher | Biomedical AI, Multimodal Health Data & Medical Time-Series

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Research Profile

PhD researcher in Information and Communications Technology working at the intersection of artificial intelligence, biomedical data science, medical time-series analysis, and multimodal health data. My research develops machine- and deep-learning methods for heterogeneous biomedical data, spanning continuous physiological monitoring, free-living wearable sensing, speech-based neurological assessment, and medical imaging. A recurring focus of my work is learning robust representations from complex biomedical data across individuals, temporal contexts, and sensing modalities through temporal modelling, multimodal fusion, subject-independent evaluation, and clinically meaningful analysis. My current research direction extends toward integrative biomedical AI that combines longitudinal physiological, clinical, imaging, and other complementary biomedical information for disease characterization, risk modelling, and personalized health applications.

Selected Research Contributions

Biomedical AI and multimodal health-data modelling. Developed machine- and deep-learning pipelines for heterogeneous biomedical information across physiological time-series, wearable sensing, speech/audio, and medical imaging. Current work includes multimodal biomedical-data analysis for AI-assisted lung-cancer screening, while previous research has addressed neurological and cardiometabolic applications.

Physiological time-series and longitudinal modelling. Developed temporal deep-learning methods for glucose prediction and meal-event detection from continuous glucose monitoring and insulin-related data, with emphasis on sequence learning, subject-independent evaluation, real-world temporal constraints, and clinically meaningful event-level analysis.

Multimodal neurological and wearable sensing. Developed speech-based neurological assessment methods using WavLM, Whisper-derived representations, LoRA/PEFT, task conditioning, and multimodal fusion for Parkinson's disease and multiple sclerosis. During a research visit at KU Leuven, developed machine-learning pipelines for free-living wrist-IMU sensing and cross-participant event detection.

Research Experience

10/2026 – Present **Post-Lauream Research Scholarship Holder**, *University of Calabria, DIMES*, Rende, Italy

Developing AI-based models for automatic tumor-lesion identification from biomedical images within the PNRR project Advancing Lung Cancer Screening: Artificial Intelligence, Multimodal Imaging and Cutting-Edge Technologies for Early Detection and Characterization. Contributing to model implementation, experimental validation, and multimodal biomedical-data analysis, with particular interest in robust and clinically applicable AI methods.

10/2024–04/2025 **International Scholar**, *KU Leuven, e-Media Research Lab, ESAT-STADIUS @ Groep T*, Leuven, Belgium

Developed machine-learning pipelines for real-world wearable sensing using wrist-mounted IMU data, including signal preprocessing, temporal alignment, windowing, annotation handling, event detection, and subject-level evaluation. Investigated eating-event detection from noisy inertial signals collected under free-living conditions and evaluated generalization across participants.

2023–2026 **PhD Researcher**, *University of Calabria, DIMES*, Rende, Italy

Developed AI methods for longitudinal physiological and multimodal health data, including CGM-based glucose prediction, meal-event detection, insulin-related temporal modelling, wearable sensing, and speech-based neurological assessment. Designed end-to-end research pipelines involving signal preprocessing, temporal representation learning, multimodal fusion, event decoding, and subject-independent validation. Led experimental analysis, evaluation design, scientific interpretation, and preparation of peer-reviewed research outputs.

06/2020–05/2023 **Research Assistant**, *Artificial Intelligence & Data Analytics Lab, Prince Sultan University*, Riyadh, Saudi Arabia, Remote

Conducted machine- and deep-learning research for healthcare applications including physiological monitoring, facial-expression analysis, medical imaging, COVID-19 analytics, and IoT-based healthcare. Contributed to preprocessing, feature extraction, model development, quantitative evaluation, and peer-reviewed publications.

Education

2023–2026 **Ph.D. in Information and Communications Technology**, *University of Calabria, DIMES*, Rende, Italy

Thesis: AI Methods for Physiological Time-Series Analysis in Healthcare: CGM Analytics and Speech-Based Neurological Assessment

2016–2019 **M.S. in Computer Science**, *International Islamic University Islamabad*, Islamabad, Pakistan

Thesis: Facial Expression Recognition Using Patch-Based Multiple LBP Descriptors in Unconstrained Environments

2009–2013 **B.S. in Computer Science**, *University of Peshawar*, Peshawar, Pakistan

Publications

Journal Papers

M. Kashif, C. Wang, S. Flesca, and P. Veltri. AI-Based Detection and Prediction of Eating Events Using Continuous Glucose Monitoring: A Comprehensive Review. *Biomed. Signal Process. Control*, vol. 127, article no. 111207, 2026.

M. Kashif, A. Hussain, A. Munir, A. B. Siddiqui, A. Abbasi, et al. A Machine Learning Approach for Expression Detection in Healthcare Monitoring Systems. *Computers, Materials & Continua*, vol. 67, no. 2, pp. 2123-2139, 2021. DOI: 10.32604/cmc.2021.014782

A. Rehman, T. Saba, **M. Kashif**, S. M. Fati, S. A. Bahaj, and H. Chaudhry, A revisit of Internet of Things technologies for monitoring and control strategies in smart agriculture, Agronomy (Basel), vol. 12, no. 1, p. 127, 2022, doi: 10.3390/agronomy12010127.

A. R. Khan, **M. Kashif**, R. H. Jhaveri, R. Raut, T. Saba, and S. A. Bahaj, Deep learning for intrusion detection and security of Internet of Things (IoT): Current analysis, challenges, and possible solutions, Secur. Commun. Netw., vol. 2022, pp. 113, 2022, doi: 10.1155/2022/4016073.

Conference Papers

M. Kashif, C. Wang, S. Flesca, and P. Veltri. Meal Episode Detection from Continuous Glucose Monitoring Using a BiLSTM Sliding-Window and Constraint-Based Decoding. *34th European Signal Processing Conference (EUSIPCO)*, 2026, pp. 856-860.

M. Kashif, P. Vizza, G. Tradigo, and S. Flesca, Task-conditioned PEFT with prompt-aligned ASR error structure fusion for Parkinson's speech assessment, in *Proceedings of the 17th ACM International Conference on Bioinformatics, Computational Biology and Health Informatics*, 2026, pp. 1-2.

M. Kashif et al., Using LSTM-based model on vocal signal analysis for the classification of multiple sclerosis, in *Lecture Notes in Computer Science,* Cham: Springer Nature Switzerland, 2025, pp. 192-196.

M. Kashif, S. Flesca, and P. Veltri. GLSTM: Using LSTM for Glucose Level Prediction. In pHealth 2024: *Proceedings of the 20th International Conference on Wearable Micro and Nano Technologies for Personalized Health - Rende, Italy*, IOS Press, 2024, pp. 103-107.

T. Saba, **M. Kashif**, and E. Afzal, Facial Expression Recognition Using Patch-Based LBPs in an Unconstrained Environment, *In 2021 1st International Conference on Artificial Intelligence and Data Analytics (CAIDA)*, 2021, pp. 105-108. DOI: 10.1109/CAIDA51941.2021.9425309.

A. Rehman, **M. Kashif**, et al. Lung Cancer Detection and Classification from Chest CT Scans Using Machine Learning Technique, *In 2021 1st International Conference on Artificial Intelligence and Data Analytics (CAIDA)*, 2021, pp. 101-104. DOI: 10.1109/CAIDA51941.2021.9425269.

Book Chapters

T. Saba, **M. Kashif**, et al. Lung Cancer Detection and Diagnosis with Deep Learning Models Evaluation. *In Artificial Intelligence and Internet of Things, CRC Press*, pp. 167-188. DOI: 10.1201/9781003097204.

M. Kashif, et al. Breast Cancer Detection and Diagnostic with Convolutional Neural Networks. *In Artificial Intelligence and Internet of Things, CRC Press*, pp. 65-84. DOI: 10.1201/9781003097204.

M. Kashif, et al. COVID-19 Outbreak Prediction After Lockdown, Based on Current Data Analytics. *In Intelligent Computing Applications for COVID-19, CRC Press*, pp. 151-168. DOI: 10.1201/9781003141105.

Technical Skills

Programming & AI: Python, PyTorch, TensorFlow/Keras, scikit-learn, NumPy, Pandas, SciPy, Git, Linux

Biomedical Data: Physiological time-series, CGM, IMU/wearable sensing, speech/audio, medical imaging, multimodal biomedical data

Methods: Temporal deep learning, LSTM/BiLSTM/GRU, Conv1D, representation learning, multimodal fusion, WavLM, Whisper, LoRA/PEFT, feature engineering, statistical modelling

Validation & Analysis: Subject-independent validation, cross-validation, event-level evaluation, ROC-AUC, sensitivity/specificity, precision/recall/F1, MAE/RMSE, reproducible ML workflows

Selected Advanced Training

2026 **International Summer School on Wearable Sensors**, *Università Campus Bio-Medico di Roma*, Rome, Italy

2026 **ELLIS Winter School on Foundation Models**, *University of Amsterdam, Amsterdam, The Netherlands*

2023–2025 **Data Science Summer Schools**, *Hertie School, Berlin, Germany*
Data science, statistical learning, machine learning, and applied computational methods.

Awards & Competitive Funding

2026 **Post-Lauream Research Scholarship**, University of Calabria, Italy

2026 **Best Poster Award**, *ACM BCB*, University of Calabria, Italy

2023–2026 **PNRR FAIR Doctoral Research Funding**, *FAIR – Future Artificial Intelligence Research Project, PNRR*, University of Calabria, Italy

2024–2025 **International Research Mobility**, KU Leuven, Belgium